

OSS Club Timeline – Technical Overview (of react-ui)

Purpose

To render dynamic, modular timelines for OSS Club's **Events** and **Milestones** using React and Vite, enabling fast development, clean component separation, and future scalability.

Architecture

1. React + Vite Stack

- **Vite** handles fast bundling, hot module replacement (HMR), and modern ES module support.
- **React** powers the component-based UI, enabling reusable logic and declarative rendering.

2. Component Modularity

- Each timeline is broken into scoped components:
 - EventCard, EventsList, EventDetailsModal, EventsTimeline
 - MilestonesTimeline
- Components are grouped by domain (events/, milestones/) for clarity and maintainability.

3. Data-Driven Rendering

- Timeline entries are loaded from static .json files:
 - eventsData.json
 - milestonesData.json
- Components receive data via props and render dynamically using .map().

4. Scoped Styling

- Each timeline has its own CSS file (EventsTimeline.css, MilestonesTimeline.css)
- Styles are modular and avoid global conflicts

5. Conditional Logic

- Optional fields like registration links or modals are rendered only when present
- `{event.link && <a href={event.link}>Register</a>}`

6. Folder Structure Discipline

- `src/components/` holds domain-specific UI logic

- `src/data/` holds structured JSON for rendering
 - `src/assets/` is reserved for images and static resources
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Component Flow

Events:

- `EventsTimeline.jsx` receives `eventsData.json`
- Renders a list of `EventCards` via `EventsList`
- `EventDetailsModal` handles expanded view (optional)

Milestones:

- `MilestonesTimeline.jsx` receives `milestonesData.json`
 - Renders each milestone with date, title, and description
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Dev Experience

- `npm run dev` launches Vite's dev server at `localhost:5173`
 - Instant HMR for rapid iteration
 - Clean separation of concerns for easy debugging and onboarding
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Future-Ready Design

- Easily extendable to:
 - API integration for dynamic data
 - Routing (`react-router-dom`) for `/events`, `/milestones`
 - Contributor dashboard, filters, and search
 - Accessibility enhancements (ARIA roles, keyboard nav)
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Current Folder Structure:

```
react-ui/
├─ node_modules/           # Installed dependencies
├─ public/
│   └─ index.html          # HTML entry point
├─ src/
│   └─ assets/             # Static assets (images, icons, etc.)
│       └─ [optional images]
│
│   └─ components/
│       └─ events/
│           ├── EventCard.jsx
│           ├── EventsList.jsx
│           ├── EventDetailsModal.jsx
│           ├── EventsTimeline.jsx
│           └─ EventsTimeline.css
│       └─ milestones/
│           ├── MilestonesTimeline.jsx
│           └─ MilestonesTimeline.css
│
│   └─ data/
│       ├── eventsData.json
│       └─ milestonesData.json
│
│   └─ App.css             # Global styles
│   └─ App.jsx             # Main app component
│   └─ index.css           # Base styles
│   └─ main.jsx            # React entry point
│
├─ .gitignore              # Git exclusions
├─ eslint.config.js        # Linting rules
├─ index.html              # Vite fallback (optional)
├─ package-lock.json       # Dependency lock file
├─ package.json            # Project metadata and scripts
├─ README.md               # Project documentation
└─ vite.config.js          # Vite build configuration
```